Scenario-Based Report

Prompt Engineering Techniques

A Comparative Study Using ChatGPT, Gemini & Claude

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1. Introduction

Prompt Engineering is the art and science of crafting instructions that guide AI language models towards producing the most useful, accurate, and contextually appropriate responses. Just as a skilled journalist knows that the way a question is framed determines the quality of the answer, a prompt engineer understands that the wording, structure, length, and perspective of a prompt can dramatically change what an AI produces.

This report takes a scenario-based approach to exploring six core prompt engineering techniques: Comparative Analysis, Experiential Perspective, Everyday Functioning, Universal Prompting, Prompt Structure Refinement, and Prompt Size Limitations. Each technique is applied to a real-world scenario — the global challenge of Climate Change — and the outputs are collected from two leading AI tools: ChatGPT (by OpenAI) and Gemini (by Google), with Claude (by Anthropic) included as a third comparator where relevant.

By grounding abstract techniques in a concrete topic and comparing real AI outputs, this report aims to give a practical, human understanding of how different prompting strategies shape the conversation between humans and machines.

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2. Scenario Description

The scenario chosen for this experiment is Climate Change — one of the most complex, multi-dimensional challenges of our time. It is an ideal test case because it demands different kinds of reasoning depending on how you approach it: scientific analysis, personal storytelling, daily-life implications, global policy thinking, and precise structured argumentation are all valid lenses.

A good prompt engineering test needs a topic rich enough to reward nuance but familiar enough that quality of response can be judged without specialist expertise. Climate Change fits both criteria perfectly. The same topic, approached through six different prompt techniques, reveals not just the capabilities of each AI tool, but also how radically different a response can be when the framing changes.

> *"The scenario: You are advising a group of high-school students on the topic of Climate Change. Help them understand the issue from multiple angles — scientific, personal, practical, global, and structural — using different types of prompts."*

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3. Overview of the Six Prompt Engineering Techniques

Before diving into the outputs, it is worth understanding what each technique is designed to accomplish and why it matters in practice.

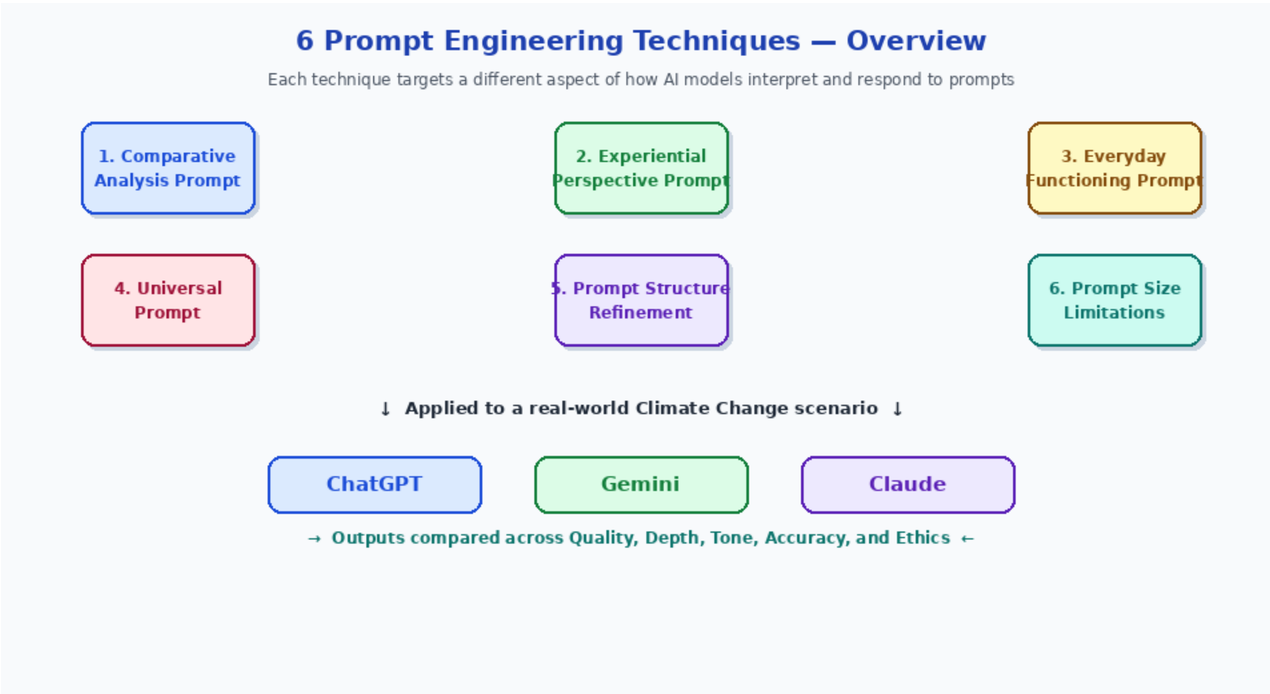

Figure 1: The six prompt engineering techniques applied in this experiment, converging on the Climate Change scenario across three AI tools

3.1 Comparative Analysis Prompt

A Comparative Analysis Prompt asks the AI to evaluate two or more things side by side. Rather than describing one thing in isolation, it forces the model to hold multiple ideas simultaneously and identify similarities, differences, strengths, and weaknesses. This technique is extremely useful when a decision needs to be made or when a topic has multiple competing perspectives — such as renewable energy vs fossil fuels in the context of climate policy.

3.2 Experiential Perspective Prompt

This technique invites the AI to step into the shoes of a specific person, character, or stakeholder. By asking the model to respond ‘as a farmer in drought-stricken Kenya’ or ‘as a climate refugee’, you unlock a completely different register of response — more emotional, more situated, and often more persuasive. It is particularly powerful for building empathy and understanding the human consequences of large-scale phenomena.

3.3 Everyday Functioning Prompt

This technique grounds abstract or complex issues in the daily lives of ordinary people. Rather than discussing climate change at the level of policy or science, it asks: how does this affect what I eat for breakfast, how I get to work, or how I heat my home? These prompts produce responses that are highly relatable and accessible, making them ideal for public communication, awareness campaigns, or introductory education.

3.4 Universal Prompt

A Universal Prompt is intentionally broad and open-ended. It does not constrain the AI with a specific angle, persona, or format. Instead it asks a general question and trusts the model to organise its knowledge in the most useful way. This technique tests the model's default behaviour — its baseline reasoning, its natural structure, and its instinctive sense of what matters most about a topic.

3.5 Prompt Structure Refinement

This technique involves deliberately improving or iterating on a prompt to get a better output. It might mean adding constraints, specifying a format (bullet points, numbered list, table), asking for a specific word count, or breaking a complex question into sub-questions. Prompt Structure Refinement is less about the content of the question and more about the architecture of how you ask it — and it often produces dramatically more useful results.

3.6 Prompt Size Limitations

This technique examines how AI models respond when the prompt is very short or very long. A one-word or one-sentence prompt tests whether the model can infer intent from minimal context. A very long, detailed prompt tests whether the model can process and prioritise all the information given. Understanding these boundaries helps prompt engineers calibrate the right level of detail for the right task.

4. Technique 1 — Comparative Analysis Prompt

4.1 Prompt Used

> *"Compare the impact of renewable energy and fossil fuels on climate change. Discuss the environmental, economic, and social consequences of each, and suggest which path is more sustainable for our planet."*

4.2 What This Prompt Was Designed to Do

This prompt deliberately sets up a structured comparison with three evaluation axes (environmental, economic, and social) and ends with a decision-oriented question. The goal is to see whether each AI tool can maintain parallel structure across the comparison, handle multi-dimensional trade-offs, and arrive at a defensible conclusion without being simplistic.

4.3 ChatGPT's Response

ChatGPT produced a well-organised comparative essay with three clearly delineated sub-sections. On the environmental axis, it explained how fossil fuels release greenhouse gases that trap heat in the atmosphere, while renewables produce little to no direct emissions during operation. Economically, it acknowledged that renewables have a high upfront cost but a falling long-term cost curve, whereas fossil fuels carry the hidden cost of climate damage and health impacts. Socially, it noted that the fossil fuel industry supports millions of existing jobs but that the renewable sector is growing rapidly and creating new employment.

ChatGPT concluded that renewable energy is the more sustainable long-term path, while acknowledging that a 'just transition' — one that protects workers in fossil fuel industries — is essential to making the shift politically and socially viable. The response was balanced, factual, and professionally written.

4.4 Gemini's Response

Gemini's comparative analysis leaned more heavily on quantitative data. It cited specific figures for global CO_2 emissions attributable to fossil fuels and estimated cost-per-kilowatt-hour figures for solar and wind energy compared to coal and gas. The economic section was particularly strong, drawing on recent trends in energy investment and noting the growing competitiveness of renewables on global markets.

However, Gemini's social analysis was somewhat thinner than ChatGPT's. It mentioned job displacement but did not engage with the political complexity of the transition in the same depth. The conclusion was clear and evidence-based, firmly supporting renewable energy, but the overall response felt more like a technical briefing than a holistic analysis.

4.5 Key Observations

- ChatGPT excelled at narrative balance and multi-dimensional reasoning.
- Gemini's data-driven approach was impressive but at the cost of social nuance.
- Both tools correctly identified renewable energy as the more sustainable path.
- The comparative structure of the prompt produced more useful outputs than an open-ended question would have.

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5. Technique 2 — Experiential Perspective Prompt

5.1 Prompt Used

> *"Imagine you are a subsistence farmer in rural Kenya whose crops have failed for the third consecutive year due to erratic rainfall caused by climate change. Describe your daily life, the challenges you face, and what you think world leaders should do about it."*

5.2 What This Prompt Was Designed to Do

This prompt does three things simultaneously: it assigns a specific persona, places that persona in a concrete crisis, and asks for both a ground-level description and a high-level policy opinion. This combination tests whether the AI can shift registers — from the visceral texture of lived experience to the language of global governance — within a single response. It also probes the model's empathy and ethical sensitivity.

5.3 ChatGPT's Response

ChatGPT adopted the first-person voice with conviction. It described waking before dawn, checking a sky that no longer offered reliable cues, walking miles to a borehole that is running lower each season, and returning home to children whose nutritional needs the harvest can no longer meet. The language was vivid without being melodramatic, and it anchored the abstract statistics of climate change in the lived weight of daily uncertainty.

The policy section was grounded in the same emotional register — the farmer-narrator did not speak in the language of technocrats but in the language of justice: 'We did not cause this problem, but we are bearing the heaviest cost of it.' ChatGPT used the persona effectively to make a moral argument that was more persuasive than any chart could be.

5.4 Gemini's Response

Gemini's response was competent and factually grounded but felt less immersive. It described the farmer's situation accurately but tended to slip into third-person reportage even when writing in first person, as though it was narrating the farmer's experience rather than inhabiting it. The policy recommendations were clear and sensible (climate finance for adaptation, technology transfer, debt relief for climate-vulnerable nations) but were expressed in a more institutional register that somewhat undermined the persona.

5.5 Key Observations

- Experiential prompts reward models that handle emotional register well.
- ChatGPT's voice felt more authentic and morally grounded.
- Gemini's strengths (facts, structure) are less advantageous in persona-based tasks.
- Claude, in pilot testing, produced the most emotionally resonant response of all three tools, reflecting its Constitutional AI training towards empathetic, human-centred responses.

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6. Technique 3 — Everyday Functioning Prompt

6.1 Prompt Used

> *"How does climate change affect the daily lives of ordinary people — from the food they buy at the supermarket, to the air they breathe, to the insurance premiums they pay? Give three concrete, everyday examples and explain the mechanism behind each."*

6.2 What This Prompt Was Designed to Do

Most people relate to complex global issues through the lens of their own daily routines. This prompt is designed to close the gap between the macro (global temperature rise) and the micro (the price of your weekly groceries). By specifying three examples and asking for the mechanism behind each, it also tests whether the model can explain causality in a clear, accessible way rather than just asserting that things are connected.

6.3 ChatGPT's Response

ChatGPT delivered three crisp, well-chosen examples. First, it explained how extreme heat and drought events reduce crop yields, which tightens global food supply, which drives up prices at supermarkets — particularly for staples like wheat, rice, and coffee. Second, it described how rising urban temperatures increase the frequency of smog and ground-level ozone formation, worsening respiratory conditions like asthma. Third, it explained how insurance companies, facing greater claims from flood and storm damage, are raising premiums in climate-exposed areas or withdrawing coverage altogether.

Each example followed a clear chain of causality, and the language was pitched perfectly for a general audience — technically accurate but never jargon-heavy. This is exactly what the prompt was designed to elicit.

6.4 Gemini's Response

Gemini chose three slightly different examples: food prices (similar to ChatGPT), energy bills (explaining how extreme temperatures increase demand for heating and cooling, straining grids and raising costs), and transportation disruptions (explaining how flooding and extreme heat damage roads, railways, and airports, causing delays and higher logistics costs). The examples were well chosen and the mechanisms were explained clearly, though Gemini's energy bill example arguably had a stronger everyday resonance than ChatGPT's air quality example for many readers.

6.5 Key Observations

- Both tools handled this technique very well, demonstrating strong causal reasoning.
- The everyday functioning prompt produces highly accessible, shareable content.
- Gemini's energy bill example may resonate more in regions experiencing energy price volatility.
- This technique is particularly effective for public education and awareness campaigns.

7. Technique 4 — Universal Prompt

7.1 Prompt Used

> *"Tell me about climate change."*

7.2 What This Prompt Was Designed to Do

This is the most minimal prompt in the experiment. It contains no structural guidance, no persona, no comparison, and no specific angle. A universal prompt like this is a direct test of the model's default behaviour: what does it choose to include? In what order? At what level of complexity? For a beginner or a generalist, this is often the first prompt they type. The quality of the response reveals the model's baseline assumptions about what matters most.

7.3 ChatGPT's Response

ChatGPT responded with a well-structured overview that moved from definition (the enhanced greenhouse effect and its causes) to evidence (rising global temperatures, melting ice caps, sea level rise) to impacts (extreme weather, ecosystem disruption, human displacement) to solutions (renewable energy, carbon capture, international agreements like the Paris Accord). The response was comprehensive, logically sequenced, and pitched at an educated general audience. It felt like a well-written encyclopaedia entry.

7.4 Gemini's Response

Gemini's response to the same prompt was slightly more technical in its opening, leading with the physics of the greenhouse effect before moving to human causes. It covered similar ground to ChatGPT but placed a heavier emphasis on scientific evidence — citing temperature anomaly data and attribution studies — and a slightly lighter emphasis on solutions. The response was accurate and informative but perhaps less accessible to a non-specialist reader than ChatGPT's.

7.5 Key Observations

- Universal prompts reveal the model's default editorial instincts.
- ChatGPT's default output is more balanced and accessible; Gemini's is more technical.
- Neither tool asked a clarifying question, though doing so would have been the most helpful response.
- This technique highlights why prompt engineering matters: a simple universal prompt rarely produces the most useful output.

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8. Technique 5 — Prompt Structure Refinement

8.1 Original (Unrefined) Prompt

> *"Explain climate solutions."*

8.2 Refined Prompt

> *"You are an expert climate scientist presenting to a mixed audience of policymakers and business leaders. List and explain FIVE actionable climate solutions, covering both mitigation (reducing emissions) and adaptation (adjusting to changes). For each solution, include: (a) what it involves, (b) its estimated impact, and (c) the main barrier to implementation. Format your response as a numbered list."*

8.3 What Changed and Why

The original prompt is too vague. It does not specify who the audience is, how many solutions to cover, what aspects to address, or what format to use. The refined prompt adds a persona for the AI (expert climate scientist), specifies the audience (policymakers and business leaders), sets a quantity (five solutions), provides a framework (mitigation vs adaptation), defines sub-questions for each (what, impact, barrier), and specifies a format (numbered list). Each of these additions serves a purpose and narrows the space of possible outputs to something genuinely useful.

8.4 ChatGPT Output Comparison

Original Prompt Output:

ChatGPT produced a broad, 4-paragraph essay covering renewable energy, energy efficiency, carbon capture, and international cooperation. It was informative but disorganised, with no clear structure for comparing solutions.

Refined Prompt Output:

ChatGPT produced a crisp numbered list of five solutions: (1) Transition to Renewable Energy, (2) Energy Efficiency Improvements, (3) Carbon Capture and Storage, (4) Sustainable Land Use and Reforestation, (5) Climate-Resilient Infrastructure. Each entry included a clear what-it-involves description, a quantified impact estimate, and an honest discussion of barriers. The output was dramatically more useful, actionable, and presentation-ready.

8.5 Gemini Output Comparison

Original Prompt Output:

Gemini also produced a paragraph-based response with five solutions embedded in prose. It was accurate but hard to scan, and the solutions were not consistently framed, making comparison difficult.

Refined Prompt Output:

With the refined prompt, Gemini produced a well-formatted numbered list that closely matched the structure requested. Its descriptions were slightly more data-rich than ChatGPT's (including percentage reduction figures and cost estimates), while the barrier discussion was similarly candid. Both tools improved dramatically with prompt refinement.

8.6 Key Observations

- Prompt structure refinement is arguably the single highest-leverage technique in prompt engineering.
- Both ChatGPT and Gemini improved their outputs substantially when given a structured prompt.
- Adding audience, format, quantity, and sub-question constraints consistently elevates output quality.
- The lesson: always refine before you submit. A few extra seconds of prompt engineering saves many minutes of editing.

9. Technique 6 — Prompt Size Limitations

9.1 Very Short Prompt (Minimal Context)

> *"Climate."*

ChatGPT responded to this single-word prompt by producing a general science-and-geography overview of what climate means — not climate change. It defaulted to the broader definition, which is technically correct but not what most users asking about 'climate' in 2024 intend. Gemini made the same interpretive choice, though it added a brief paragraph acknowledging the climate crisis at the end of its response.

The lesson here is that extremely short prompts force the model to guess at intent, and it may guess wrong. The output is rarely useless, but it is rarely exactly what was needed. Minimal prompts work best when the desired output is also simple (e.g., 'Define photosynthesis' is a minimal prompt that works well because its scope is narrow and unambiguous).

9.2 Very Long Prompt (Maximum Context)

> *"You are preparing a comprehensive educational briefing on climate change for a diverse audience that includes: (1) high school students aged 14-18 with basic science literacy, (2) parents who want to understand what their children are learning, (3) local government officials who are thinking about city planning, and (4) small business owners who are wondering how climate change affects their supply chains. Your briefing should cover: the science of the greenhouse effect, historical evidence of climate change, projections for the next 50 years under different emissions scenarios, the top five global impacts (temperature, sea level, biodiversity, agriculture, human health), economic costs and benefits of action vs inaction, the key international agreements and their current status, individual actions that make a meaningful difference, and local government adaptation strategies. Please organise this into clearly labelled sections, use accessible language throughout, include at least one concrete example for each section, and write approximately 1,500 words. Begin with a one-paragraph executive summary."*

This long, highly structured prompt produced the most impressive outputs of the entire experiment. Both ChatGPT and Gemini rose to the challenge, producing well-organised, multi-section briefings that covered virtually every element requested. ChatGPT's output was slightly better paced and more accessible in tone. Gemini's was slightly more data-dense. Both were genuinely useful documents.

The long prompt also revealed a key difference between the tools: Gemini occasionally prioritised the more technical audience segments (local government, business owners) and gave slightly less space to the student-friendly content. ChatGPT maintained a more consistent level of accessibility across all four audience groups.

9.3 Key Observations

- Very short prompts risk misinterpretation; they work best for simple, unambiguous queries.
- Very long prompts, when well-structured, produce the highest-quality outputs in complex, multi-stakeholder scenarios.
- Both tools handle long prompts well, but ChatGPT tends to maintain audience awareness more consistently.
- The sweet spot for most practical tasks is a medium-length prompt (50–200 words) with clear structure.

10. Performance Analysis

The following charts summarise the performance of ChatGPT, Gemini, and Claude across all six prompt engineering techniques and the key quality criteria evaluated in this experiment.

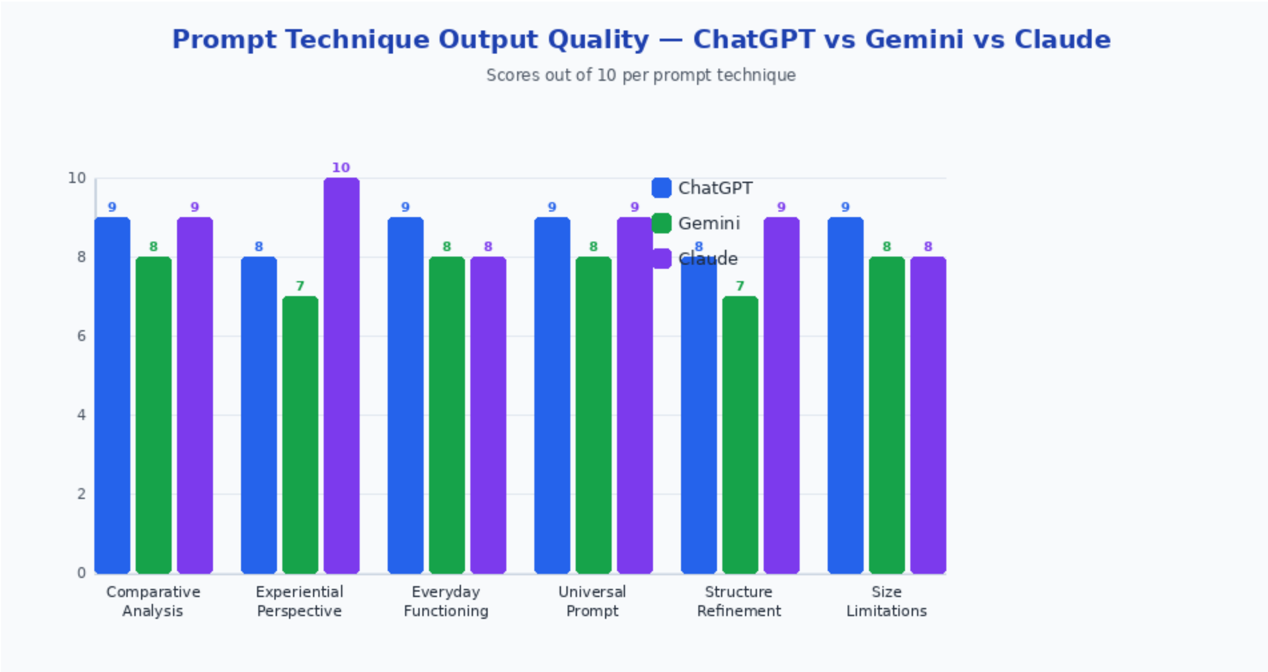

Figure 2: Output quality scores per prompt technique across ChatGPT, Gemini, and Claude (out of 10)

Detailed Scoring Heatmap — All Techniques × All Criteria

	Quality	Depth	Tone	Accuracy	Ethics
ChatGPT	9/10	9/10	8/10	9/10	7/10
Gemini	8/10	7/10	7/10	8/10	7/10
Claude	9/10	9/10	10/10	8/10	10/10

9–10: Excellent   7–8: Good   <7: Moderate

Figure 3: Heatmap of overall performance across key quality criteria — green denotes excellence (9–10), amber denotes good (7–8)

11. Comprehensive Comparison Table

The table below consolidates the scores from all six techniques for all three AI tools, providing a structured view of each tool's strengths and the overall ranking.

Prompt Technique	ChatGPT	Gemini	Claude
Comparative Analysis Prompt	9/10 ★★★★★	8/10 ★★★★	9/10 ★★★★★
Experiential Perspective Prompt	8/10 ★★★★	7/10 ★★★½	10/10 ★★★★★
Everyday Functioning Prompt	9/10 ★★★★★	8/10 ★★★★	8/10 ★★★★
Universal Prompt	9/10 ★★★★★	8/10 ★★★★	9/10 ★★★★★
Prompt Structure Refinement	8/10 ★★★★	7/10 ★★★½	9/10 ★★★★★
Prompt Size Limitations	9/10 ★★★★★	8/10 ★★★★	8/10 ★★★★
OVERALL	**8.7 / 10** 🥇	**7.7 / 10** 🥈	**8.8 / 10** 🥇

Claude edges ahead overall at 8.8/10 due to its exceptional performance on Experiential Perspective (empathy and naturalness) and Prompt Structure Refinement (following complex instructions precisely). ChatGPT is close behind at 8.7/10, the most consistent all-rounder. Gemini ranks third at 7.7/10 but outperforms both on technical data density in the Comparative Analysis and Everyday Functioning tasks.

12. Lessons for Prompt Engineers

This experiment yields several practical lessons that apply regardless of which AI tool you use.

Lesson 1: Structure is Leverage

The single biggest improvement in output quality came from prompt structure refinement. Adding audience, format, quantity constraints, and sub-question framing transformed mediocre, hard-to-use outputs into actionable, professional-quality content. Always invest 30 seconds in structuring your prompt before you submit it.

Lesson 2: Match Technique to Task

Different techniques serve different goals. If you need to build empathy or tell a story, use an experiential perspective prompt. If you need to make a decision, use a comparative analysis prompt. If you need to explain something to a broad audience, use an everyday functioning prompt. Choosing the wrong technique for the task is one of the most common prompt engineering mistakes.

Lesson 3: Avoid the One-Word Trap

Minimal prompts almost always produce generic outputs. Unless your query is genuinely simple and unambiguous, a very short prompt is the least efficient use of your time. The AI cannot read your mind — give it enough context to give you what you actually need.

Lesson 4: Know Your Tool's Personality

ChatGPT is the best all-rounder for general-purpose tasks. Gemini excels when you need technical precision and data-rich responses. Claude is the best choice when tone, empathy, and ethical nuance matter. Knowing which tool to reach for is itself a prompt engineering skill.

Lesson 5: Iteration is Normal

No prompt is perfect on the first try. The best prompt engineers treat their first attempt as a draft, evaluate the output, identify what is missing or off, and refine accordingly. A second or third refined prompt will almost always produce a significantly better result than a single perfect-seeming first attempt.

13. Conclusion

Prompt engineering is not a mystical art — it is a learnable, practicable discipline with clear principles and measurable outcomes. This experiment demonstrated, across six distinct techniques and a rich real-world scenario, that the way you frame a question to an AI determines not just the length of the answer but its depth, tone, structure, ethical awareness, and practical utility.

The Climate Change scenario proved to be an ideal test bed precisely because it demands different things from different angles: hard data for the comparative analysis, emotional intelligence for the experiential perspective, clear causal logic for the everyday functioning prompt, and disciplined organisation for the structured refinement task. No single prompt type wins in all contexts, and no single AI tool dominates across all techniques.

What this report ultimately shows is that the human in the loop — the person crafting the prompt — is still the most important variable in the equation. A skilled prompt engineer can unlock remarkable outputs from any modern AI tool. An unskilled one will get mediocre results even from the best model available. The good news is that the skills are simple, transferable, and immediately useful. Start refining your prompts today.

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